Baseline Report: RAG Sufficient Context — Selective Generation Pipeline

GitHub Repository: https://github.com/SachaYT1/rag-sufficient-context

1. Product Name

RAG Sufficient Context — a pipeline for explaining and reducing hallucinations in Retrieval-Augmented Generation systems via selective generation, inspired by the ICLR 2025 paper *"Sufficient Context: A New Lens on Systems Retrieval Augmented Generation"*.

2. Domain of Application

Reliability and calibration for Retrieval-Augmented Generation (RAG) in open-domain question answering (QA). The pipeline targets any system where an LLM generates answers based on retrieved documents — search engines, customer support bots, knowledge bases, educational assistants, and enterprise QA systems.

3. Motivation

Large Language Models integrated into RAG pipelines are increasingly deployed in production. However, a critical failure mode persists: **when retrieved context is insufficient, models tend to hallucinate rather than abstain**. This creates a trust problem — users cannot distinguish confident correct answers from confident hallucinations.

The ICLR 2025 paper by Joren et al. introduces the concept of *sufficient context* as a new diagnostic lens. We are motivated to:

- Reproduce and validate this idea in a practical, university-friendly setting
- Demonstrate that combining a sufficiency signal with model confidence leads to better answer-vs-abstain decisions
- Build a fully reproducible pipeline that others can extend

4. Real-World Problem

In real-world RAG deployments:

- **Retrieval is imperfect**: search often returns partially relevant, incomplete, or misleading documents
- **LLMs don't know when to stop**: even strong models (GPT-4, LLaMA-3) frequently generate plausible-sounding but incorrect answers when the context doesn't contain the needed information
- **Consequences are real**: in healthcare, legal, financial, and educational settings, hallucinated answers can cause harm

Our product addresses this by giving the system a mechanism to **recognize when it shouldn't answer** — reducing hallucination rates while maintaining high accuracy on questions it does answer.

5. Technical Problem

We frame the technical problem as **selective generation**: given a question Q and retrieved context C, decide whether to answer or abstain.

The challenge decomposes into:

1. **Context sufficiency classification**: Is the retrieved context C sufficient to answer Q? This is a binary classification task on (Q, C) pairs, performed *without access to ground-truth answers*.
2. **Confidence estimation**: How confident is the LLM in its generated answer? We use P(Correct)-style self-reported probability.
3. **Gating decision**: Combine sufficiency and confidence into a single score via logistic regression, and apply a threshold to decide answer vs. abstain.

The key insight from the ICLR 2025 paper is that **confidence alone is insufficient** — a model can be confidently wrong when the context is misleading. Adding the sufficiency signal provides an orthogonal dimension that captures whether the context *supports* any definitive answer.

Formal setup:

- Features: x_1 = confidence(Q, C), x_2 = s(Q, C) $\in$ {0, 1}
- Gate: $g(x_1, x_2)$ trained via logistic regression
- Decision: if $g(x_1, x_2) < \tau$ → abstain; otherwise → answer
- Metric: Selective Accuracy vs. Coverage curves

6. Baseline Implementation

6.1 Dataset

HotPotQA (dev subset, 500 examples) — a multi-hop QA benchmark requiring reasoning over multiple Wikipedia paragraphs. We use the "distractor" setting where each question comes with 10 paragraphs (2 gold + 8 distractors).

6.2 Retrieval

- **Method**: BM25 (via `rank_bm25` library) — per-question re-ranking over the HotPotQA distractor set
- **Process**: For each question, we index the 10 provided paragraphs and retrieve the top-5 most relevant
- **Context construction**: Concatenate top-5 passages, truncate to 4096-token budget

6.3 Generation

- **Model**: Mistral-7B-Instruct-v0.3 (via HuggingFace Transformers, run on Google Colab T4 GPU)
- **Prompt**: Structured prompt requesting JSON output with answer and confidence score
- **Inference**: Greedy decoding (temperature = 0), max 256 new tokens
- **Runtime**: ~77 minutes for 500 examples on T4 (~9.3s per example)

6.4 Evaluation

- **Exact Match (EM)**: Normalized string comparison (lowercase, remove articles/punctuation)
- **F1 Score**: Token-level precision/recall

- **Abstention detection**: Regex whitelist matching phrases like "I don't know", "cannot be determined"
- **Categorization**: Each response is labeled as `correct` (EM=1 or F1≥0.5), `abstain` (matches abstention pattern), or `hallucinate` (neither)

6.5 Sufficient Context Autorater

- **Approach**: Use the same Mistral-7B model with a dedicated autorater prompt
- **Chunking**: Split context into chunks of ~1400 tokens
- **Aggregation**: OR-rule — if any chunk is rated as sufficient, the full context is labeled sufficient
- **Runtime**: ~30 minutes for 500 examples on T4 (~3.6s per example)

6.6 Selective Generation Gate

- **Baseline**: Threshold on confidence only (x_1)
- **Proposed**: Logistic regression on (confidence, sufficiency) with threshold sweep
- **Evaluation**: Selective Accuracy vs. Coverage curves comparing both approaches (cross-validated predictions to avoid overfitting)

6.7 Project Structure

```
rag-sufficient-context/
├── src/
│   ├── retrieval.py      # BM25 retriever + context construction
│   ├── generation.py     # LLM prompting, answer extraction
│   ├── evaluation.py     # EM/F1, abstain detection, categorization
│   ├── autorater.py      # Sufficient context autorater
│   ├── confidence.py     # P(Correct) confidence estimation
│   ├── gate.py           # Logistic regression gate + plotting
│   └── utils.py          # Shared helpers
├── configs/default.yaml  # Hyperparameters
├── notebooks/
│   └── main_pipeline.ipynb  # Colab end-to-end notebook
├── data/                 # Cached results
└── results/              # Outputs, plots, metrics
```

6.8 Reproducibility

- Fixed random seed (42) across all experiments
- Cached retrieval results and autorater labels
- Single Colab notebook for end-to-end reproduction
- All dependencies pinned in `requirements.txt`

7. Experimental Results

7.1 Baseline Generation Performance

Metric	Value
Total examples	500

Metric	Value
Correct	201 (40.2%)
Abstain	135 (27.0%)
Hallucinate	164 (32.8%)
Mean EM	0.312
Mean F1	0.405

The model answers correctly in 40.2% of cases, but **hallucinates in 32.8%** — nearly one in three answers is a confident but wrong response. The model abstains in 27.0% of cases.

7.2 Context Sufficiency Analysis

The autorater classified **348 out of 500 (69.6%)** contexts as sufficient and **152 (30.4%)** as insufficient.

Category breakdown by sufficiency:

	Abstain	Correct	Hallucinate	Total
Insufficient	102 (67.1%)	20 (13.2%)	30 (19.7%)	152
Sufficient	33 (9.5%)	181 (52.0%)	134 (38.5%)	348

Key observations:

- **When context is insufficient**, the model mostly abstains (67.1%) — this is the desired behavior
- **When context is sufficient**, the model answers correctly 52.0% of the time, but still hallucinates 38.5%
- Only 13.2% of insufficient-context examples get correct answers (likely by chance or parametric knowledge)
- The sufficiency signal clearly separates high-accuracy vs. high-risk scenarios

7.3 Confidence Calibration Problem

Category	Mean Confidence
Correct	1.000
Hallucinate	0.998
Abstain	0.019

This reveals a critical finding: **Mistral-7B's self-reported confidence is almost completely uncalibrated**. The model reports near-perfect confidence (0.998–1.000) for both correct answers and hallucinations. Only abstentions receive low confidence. This means confidence alone cannot distinguish correct from hallucinated answers — it can only detect explicit abstentions.

This is precisely the scenario where the sufficient-context signal adds value: it provides an orthogonal dimension that confidence cannot capture.

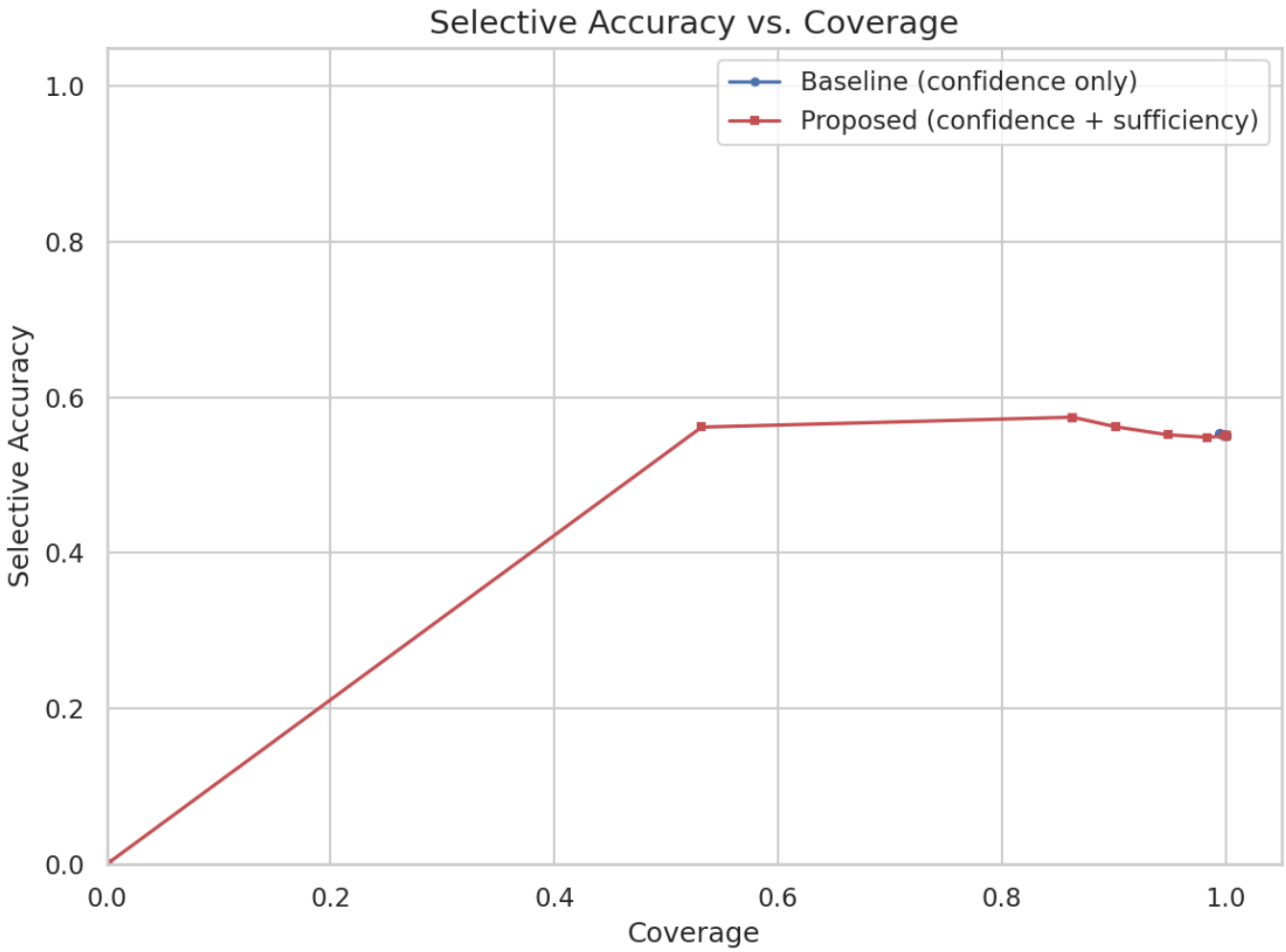
7.4 Selective Generation Gate

The logistic regression gate was trained on 365 examples (excluding abstentions) with two features:

Feature	Weight
Confidence (x_1)	0.224
Sufficiency (x_2)	**0.645**
Intercept	-0.577

The **sufficiency weight (0.645) is nearly 3x the confidence weight (0.224)**, confirming that the sufficiency signal carries more predictive power for distinguishing correct from hallucinated answers. This is consistent with the confidence calibration problem — since confidence is nearly identical for correct and hallucinated answers, the gate relies primarily on the sufficiency signal.

7.5 Selective Accuracy vs. Coverage Curves



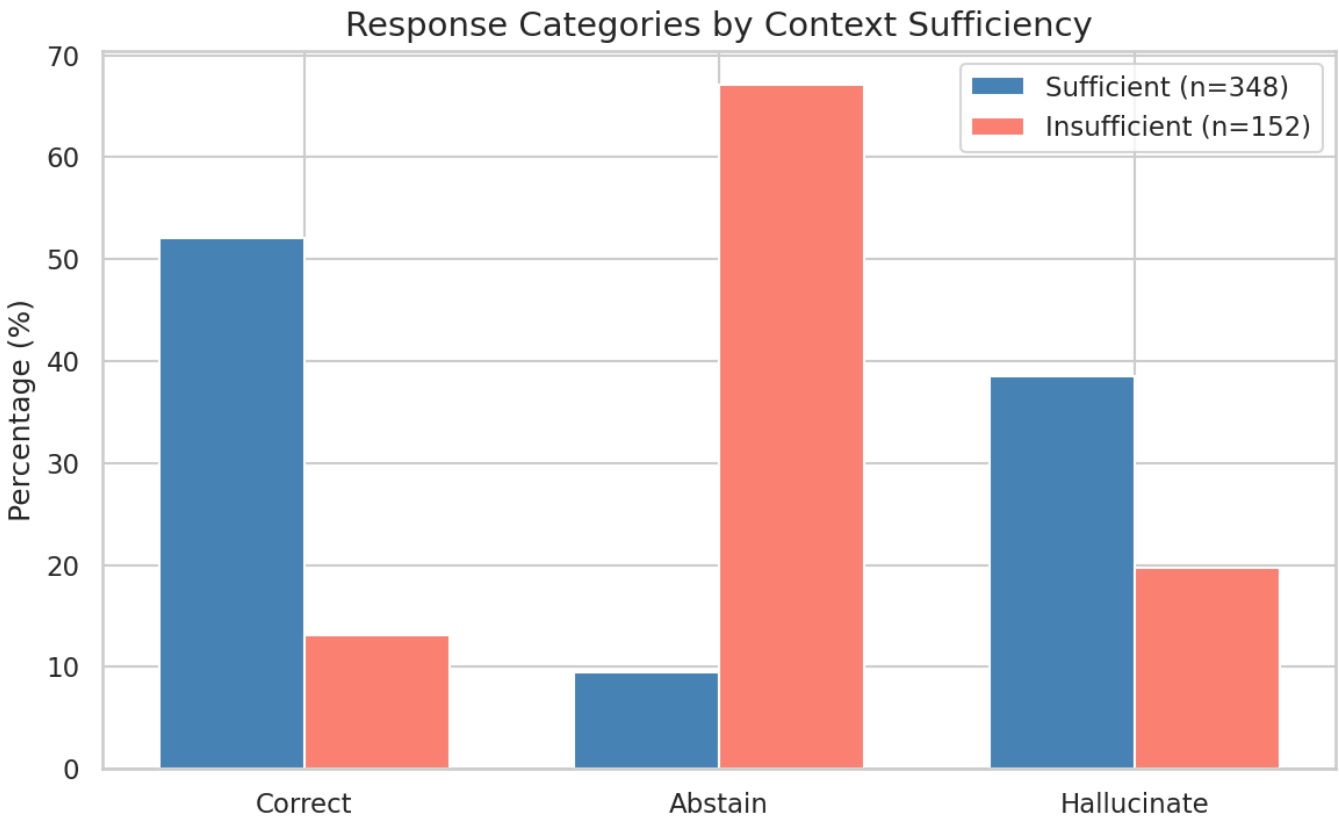
The accuracy-coverage plot shows that the baseline (confidence only) and the proposed method (confidence + sufficiency) produce **nearly overlapping curves**. Both achieve ~55% selective accuracy at full coverage, rising to ~58% at ~85% coverage.

Analysis of why the curves overlap:

The near-identical curves are explained by the confidence calibration problem (Section 7.3). Since confidence is binary (≈0 for abstentions, ≈1 for everything else), sweeping a confidence threshold produces only two operating points: either include all non-abstain examples (coverage ≈73%, accuracy ≈55%) or include nothing. The gate, despite having a better sufficiency signal, operates on the same near-binary confidence landscape and cannot leverage it for fine-grained thresholding.

This is a meaningful negative result: **P(Correct)-style self-reported confidence from Mistral-7B is too poorly calibrated to enable effective selective generation via threshold sweeping**. The gate's positive sufficiency weight (0.645) shows the signal is informative, but its utility is bottlenecked by the confidence feature's lack of granularity.

7.6 Sufficiency Breakdown Visualization



The bar chart clearly shows the diagnostic value of the sufficiency signal:

- **Sufficient contexts**: 52% correct, 9.5% abstain, 38.5% hallucinate
- **Insufficient contexts**: 13% correct, 67% abstain, 20% hallucinate

The sufficiency label alone shifts the distribution dramatically — from majority-correct to majority-abstain — validating the ICLR 2025 paper's core claim that context sufficiency is a powerful lens for understanding RAG behavior.

8. Contribution of Each Groupmate

Team Member	Role	Contribution

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Aleksandr Gavkovskii	Project Lead	End-to-end project architecture and repo setup; retrieval pipeline (`retrieval.py`); generation module (`generation.py`); evaluation framework (`evaluation.py`); sufficient context autorater (`autorater.py`); shared utilities and LLM inference abstraction (`utils.py`); integration of all modules; Colab notebook development and experiment execution; baseline report writing and results analysis; code review of all components
Ilya Maksimov	Confidence & Prompting	Confidence estimation module (`confidence.py`); prompt engineering and tuning for QA and autorater prompts; testing and debugging on Colab
Karim Zakirov	Gate & Visualization	Selective generation gate (`gate.py`); visualization and plotting (accuracy-coverage curves, sufficiency breakdown charts); manual labeling coordination

9. Lessons Learned

1. **Confidence calibration is the bottleneck**: The most striking result is that Mistral-7B's P(Correct) self-reported confidence is nearly useless for distinguishing correct from hallucinated answers (both ≈1.0). This makes the selective generation gate's threshold sweep ineffective, as there is no gradient to sweep over. Future work must address this with better confidence estimation methods (e.g., P(True) self-consistency, token-level entropy).

2. **Sufficiency signal is genuinely informative**: Despite the confidence calibration failure, the sufficiency autorater successfully separates high-risk from low-risk scenarios. When context is insufficient, the hallucination rate drops from 38.5% to 19.7%, and the abstention rate rises from 9.5% to 67.1%. The gate's learned weight confirms this: sufficiency weight (0.645) is 3x the confidence weight (0.224).

3. **Prompt engineering is critical**: Small changes in how we prompt the LLM for JSON output significantly affect parsing reliability. Constrained output formats need careful iteration.

4. **BM25 is a strong baseline**: Despite being a simple lexical method, BM25 performs surprisingly well on HotPotQA's distractor setting, where the candidate set is small (10 passages).

5. **Modular design pays off**: Splitting the pipeline into independent modules (retrieval, generation, evaluation, autorater, gate) made parallel development across three team members smooth.

6. **Reproducibility requires discipline**: Caching intermediate results (retrieval, generation, autorater) was essential — the full pipeline takes ~2 hours on Colab T4. Without caching, restarted sessions would need to re-run everything.

10. Further Goals

Technical Goals (Next Steps)

- **Fix the confidence bottleneck**: Implement P(True) self-consistency (generate N sampled answers, compute agreement) as a replacement for P(Correct). This should produce a more granular confidence signal that enables meaningful threshold sweeping.
- **Token-level entropy**: Extract per-token log-probabilities during generation and use entropy as an alternative confidence signal — this avoids relying on self-reported probabilities entirely.
- **Dense retrieval comparison**: Add e5-base-v2 as an alternative retriever and compare with BM25.
- **Second dataset**: Evaluate on MuSiQue to test generalization beyond HotPotQA.
- **Ablation studies**: Vary chunk size (800–2000 tokens), context budget (2k–8k tokens), and top-k (3–10).
- **Autorater validation**: Complete manual labeling of 80–150 examples and estimate autorater accuracy.
- **LLM-based evaluator**: Add a separate LLM judge for answer quality beyond EM/F1 (to catch partial-credit answers that EM misses).